Activity: linear mixed models

Observe data $(\mathbf{x}_{ij}, Y_{ij}) \in \mathbb{R}^{k+1} \times \mathbb{R}$ from m groups ($i = 1, ..., m$) with n_i observations ($j = 1, ..., n_i$) in the ith group, and $n = \sum_{i=1}^{m} n_i$ observations total. Let u_i and ε_{ij} be random variables with mean 0. A linear mixed model with a **random intercept** is

$$Y_{ij} = u_i + \mathbf{x}_{ij}^T \beta + \varepsilon_{ij} \tag{1}$$

It can help to write this model in matrix form:

$$\begin{bmatrix} Y_{11} \\ Y_{12} \\ \vdots \\ Y_{1n_1} \\ Y_{21} \\ \vdots \\ Y_{mn_m} \end{bmatrix} = \begin{bmatrix} \mathbf{x}_{11}^T \\ \mathbf{x}_{12}^T \\ \vdots \\ \mathbf{x}_{1n_1}^T \\ \mathbf{x}_{21}^T \\ \vdots \\ \mathbf{x}_{mn_m}^T \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 1 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ \\ \vdots \\ u_m \end{bmatrix} + \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{12} \\ \vdots \\ \varepsilon_{1n_1} \\ \varepsilon_{21} \\ \vdots \\ \varepsilon_{mn_m} \end{bmatrix}$$

Or, written more concisely:

$$\begin{aligned} &\mathbf{y} = \mathbf{X}\beta + \mathbf{Z}u + \varepsilon \\ &u \sim N(\mathbf{0}, \sigma_u^2 \mathbf{I}_m), \quad \varepsilon \sim N(\mathbf{0}, \sigma_\varepsilon^2 \mathbf{I}_n), \quad u, \varepsilon, \mathbf{X} \text{ independent} \end{aligned} \tag{2}$$

1. Show that

$$\mathbf{y}|\mathbf{X} \sim N(\mathbf{X}\beta, \Sigma)$$

where

$$\Sigma = \begin{bmatrix} \Sigma_1 & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \Sigma_2 & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \Sigma_3 & \cdots & \mathbf{0} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \cdots & \Sigma_m \end{bmatrix}$$

where each $\Sigma_i \in \mathbb{R}^{n_i \times n_i}$ is the variance-covariance matrix for group i, and

$$\Sigma_i = \begin{bmatrix} \sigma_u^2 + \sigma_\varepsilon^2 & \sigma_u^2 & \cdots & \sigma_u^2 \\ \sigma_u^2 & \sigma_u^2 + \sigma_\varepsilon^2 & \cdots & \sigma_u^2 \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_u^2 & \sigma_u^2 & \cdots & \sigma_u^2 + \sigma_\varepsilon^2 \end{bmatrix}$$